C5.01 Exothermic and endothermic reactions

Energy can be transferred during a chemical reaction. This occurs due to the change in the chemical bonds during a reaction (a change in the chemical energy of the system). It is most often observed as a heat change (the reaction feeling warmer or cooler) although other observations, such as light being given out or a sound being produced, are also possible.

Many reactions, such as combustion (burning) give out heat energy. In some reactions, energy is absorbed. In one of the most important chemical reactions on the planet, plants use the energy in sunlight to convert carbon dioxide and water into carbohydrates by the process of photosynthesis (Figure C5.03).

Figure C5.03 Photosynthesis takes place in the green leaves of plants.

Processes that release heat energy (thermal energy) to the surroundings are called exothermic processes (Figure C5.04a). Adding zinc to copper(II) sulfate solution is an exothermic chemical reaction. The heat energy released in these exothermic reactions leads to an increase in the temperature of the surroundings and the reaction will feel hot. The surroundings include:

- the reaction mixture in the test-tube
- the air around the test-tube
- the test-tube itself
- the thermometer in the test-tube.

This final point is important: it is the thermal energy (heat) transferred to the thermometer that causes the rise in temperature shown on the thermometer.

Processes that take in heat energy from the surroundings are endothermic processes (Figure C5.04b). Endothermic reactions absorb heat. The heat energy is taken in by the reacting substances (the system) and so leads to a decrease in the temperature of the surroundings. The reaction mixture will feel cold. When you try to remember these particular terms, concentrate on the first letters of the words involved:

- EXothermic means that heat EXits the reaction.
- ENdothermic means that heat ENters the reaction.

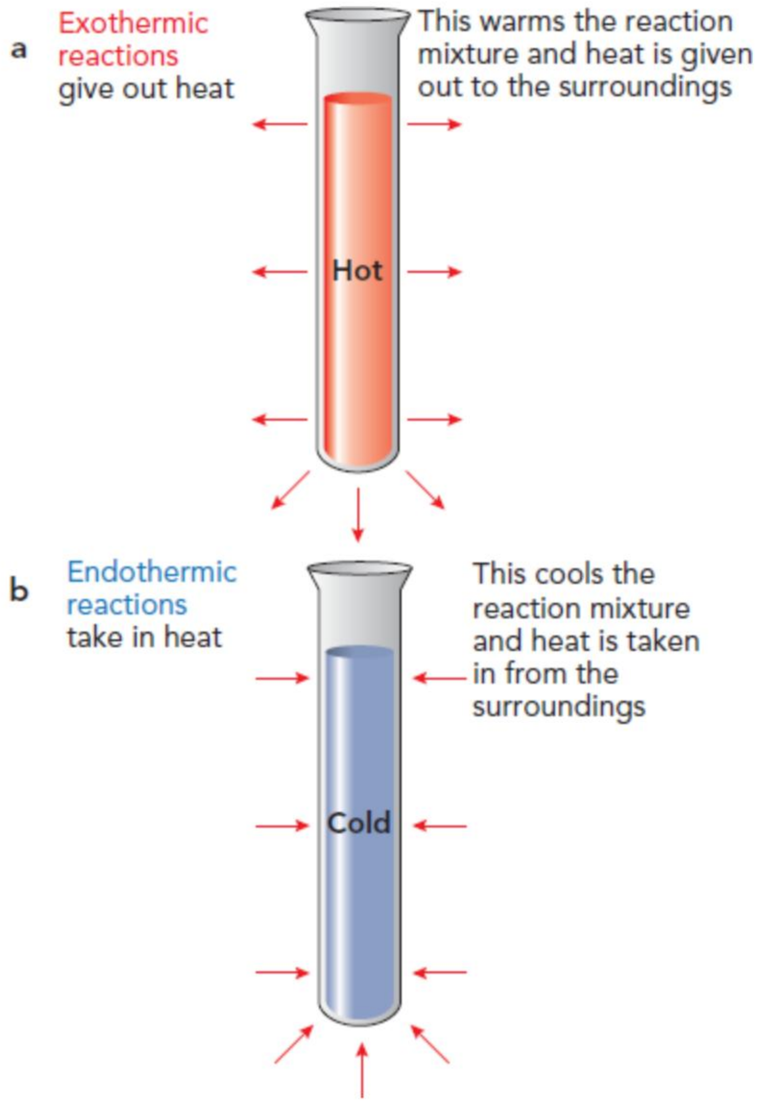

Figure C5.04 a: An exothermic reaction gives out heat energy to the surroundings. **b:** An endothermic reaction takes in thermal energy from the surroundings.

Energy level diagrams

Exothermic and endothermic changes for reactions can be shown using **energy level diagrams** (also called **reaction pathway diagrams**). Such diagrams show:

- the energy of the reactants and products on the vertical axis (*y*-axis)
- the 'reaction pathway' (progress of reaction) on the horizontal axis (*x*-axis), with reactants on the left and products on the right
- the energy change indicated by an arrow (upward or downward) between the reactants and products.

Methane is the simplest hydrocarbon molecule and the main component of natural gas. When it burns, it reacts with oxygen. The products are carbon dioxide and water vapour (water in the gaseous state):

$$\text{methane} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water}$$
$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

Figure C5.05 shows an energy level diagram for the reaction of methane and oxygen producing carbon dioxide and water.

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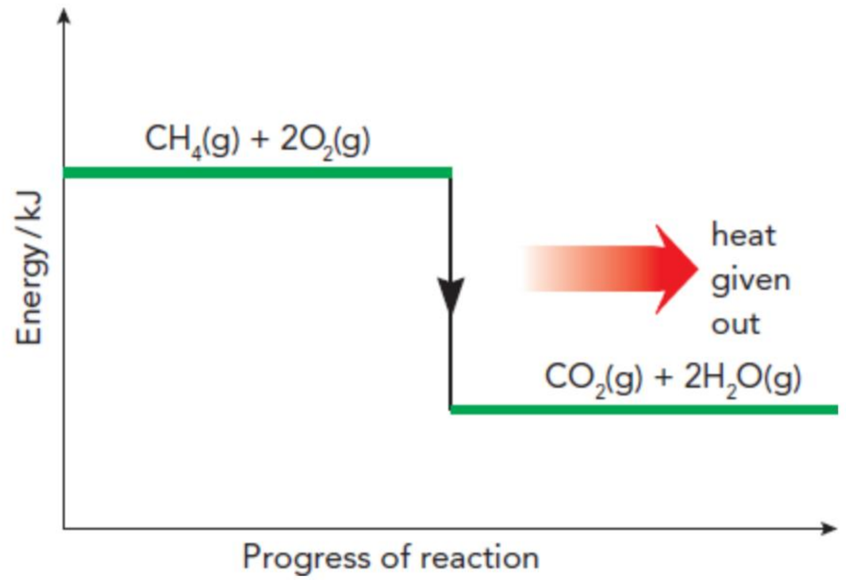


Figure C5.05 Energy level diagram for the exothermic reaction between methane and oxygen.

For an exothermic reaction you can see that:

- the energy of the reactants is higher than the energy of the products
- the black arrow points downwards to show that energy is given out (released).

The reaction between nitrogen and oxygen is endothermic. It is one of the reactions that take place when fuel is burnt in car engines.

The equation for this reaction is:

$$\begin{array}{ccccc} \text{nitrogen} & + & \text{oxygen} & \rightarrow & \text{nitrogen monoxide} \\ N_2(g) & + & O_2(g) & \rightarrow & 2\,NO(g) \end{array}$$

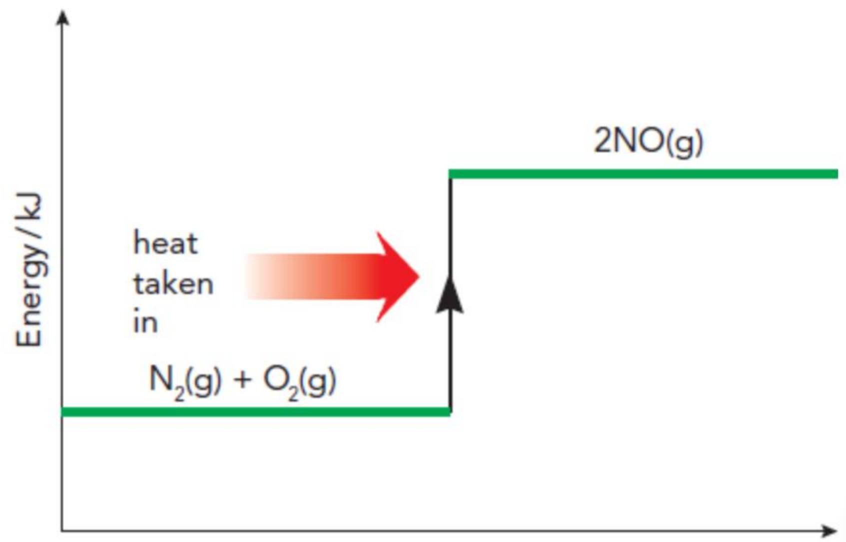


Figure C5.06 An energy level diagram for the reaction between nitrogen and oxygen.

Figure C5.06 shows an energy level diagram for this reaction.

For an endothermic reaction you can see that:

- the energy of the reactants is lower than the energy of the products
- the black arrow points upwards to show that energy is absorbed (taken in).

WORKED EXAMPLE C5.01

Magnesium burns in oxygen, giving off heat and light. Draw an energy level diagram to represent this reaction.

Step 1: Write a word equation for the reaction:

magnesium + oxygen → magnesium oxide

Step 2: Work out the formulae for the substances involved and write a balanced symbol equation for the reaction.

$$Mg + O_2 \rightarrow MgO$$

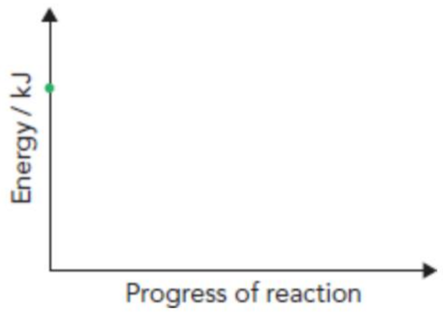
Balancing gives:

$$2Mg + O_2 \rightarrow 2MgO$$

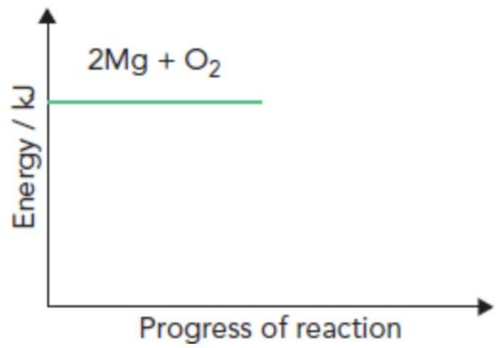
Step 3: Identify whether energy is absorbed (endothermic) or released (exothermic).

The question tells us that energy is released as heat and light.

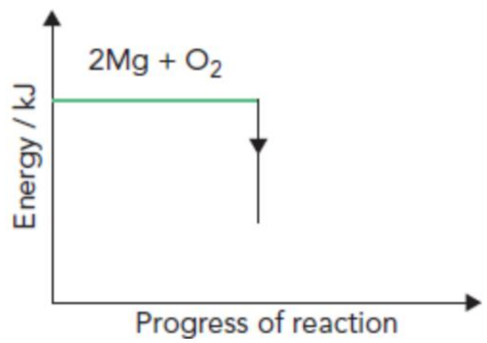
Step 4: Draw and label the axes.



Step 5: Mark the reactants by drawing a horizontal line to the left-hand side of your axis. Write the reactants half of the balanced symbol equation on this line.



Step 6: As the reaction is exothermic, draw a downwards arrow from the reactants line to represent energy being released.



Step 7: Draw a horizontal line at the end of the arrow to represent the products. Label this with the products half of the balanced symb

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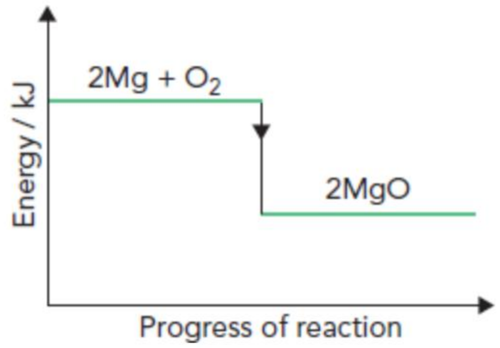


Enthalpy and enthalpy change (Δ*H*)

The thermal energy (heat) content of a system is called the **enthalpy (*H*)** of the system. The transfer of thermal energy during a reaction is called the **enthalpy change (Δ*H*)** (the symbol Δ (delta) means 'change in') of the reaction.

- When heat is released from the system to the surroundings in an exothermic reaction, the enthalpy of the system decreases and Δ*H* is negative.
- When heat is taken in (absorbed) by the system from the surroundings in an endothermic reaction, the enthalpy of the system increases and Δ*H* is positive.

These ideas fit with the direction of the black arrows shown in the energy diagrams (Figures C5.05 and C5.06). The energy given out or taken in is measured in kilojoules (kJ). One kilojoule (1 kJ) = 1000 joules (1000 J). It is usually calculated per mole of a specific reactant or product (kJ/mol). The enthalpy change (Δ*H*) for the burning of methane is high (Figure C5.07). This high value explains why methane is such a useful fuel.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

$$\Delta H = -728 \text{ kJ/mol}$$

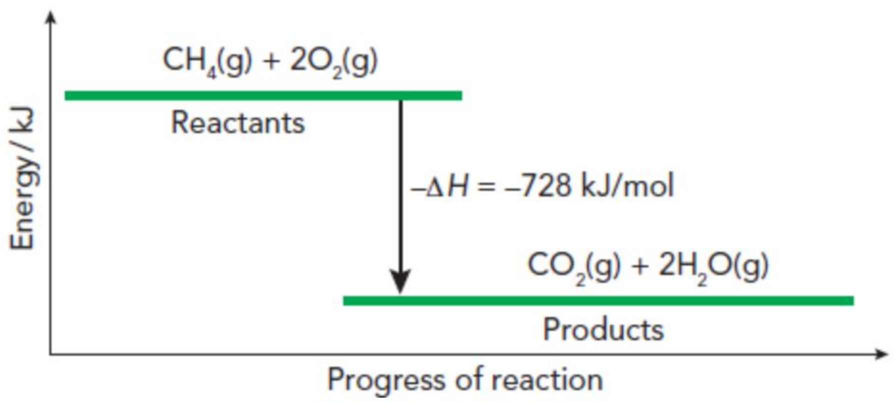


Figure C5.07 Detailed reaction pathway diagram for the exothermic reaction between methane and oxygen, showing the enthalpy change (Δ*H*).

Questions

C5.01 Which type of reaction takes in heat from its surroundings?

C5.02 **a** Draw an energy level diagram for the following reaction, which is exothermic.

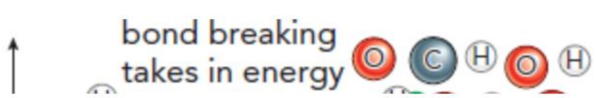
$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$

b Label the enthalpy change for the reaction and state whether this will be positive or negative.

C5.03 What key features of an energy level diagram would indicate an endothermic reaction?

Bond breaking and bond making

Most substances, apart from the noble gases (Group VIII), are involved in chemical bonding of some type (Chapter C2). During a reaction, some of the bonds between the particles are broken and new bonds are formed. During the reaction between methane and oxygen, as with all other reactions, bonds are first broken and then new bonds are made (Figure C5.08). In methane molecules, carbon atoms are covalently bonded to hydrogen atoms. In oxygen gas, the atoms are held together in diatomic molecules. During the reaction, all these bonds must be broken. As chemical bonds are forces of attraction between atoms or ions, breaking these bonds requires energy. Energy is needed to pull the atoms apart, so breaking chemical bonds takes in energy from the surroundings. Bond breaking is an endothermic process. New bonds are then formed: between carbon and oxygen to make carbon dioxide (CO_2), and between hydrogen and oxygen to form water (H_2O). Making these chemical bonds gives out energy to the surroundings. Bond making is an exothermic process.



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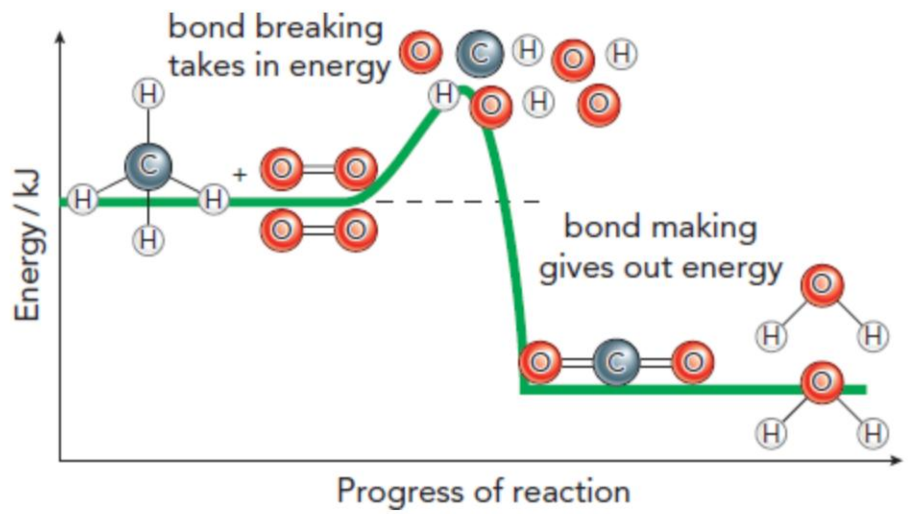

Figure C5.08 Reaction of methane with oxygen first involves the breaking of bonds in the reactants. This is followed by the formation of the new bonds of the products.

Some bonds are stronger than others. Stronger bonds have a higher **bond energy**. They require more energy to break, but they give out more energy when they are formed. In Figure C5.08 the bonds in the products are stronger than those in the reactants. This means more energy is given out when the new bonds form than is needed to break the original bonds. Overall, this reaction gives out thermal energy (heat). It is an exothermic reaction.

The relationship between making and breaking bonds and the energy involved is summarised in the memory aid 'MEXOBENDO': **M**aking bonds = **EXO**thermic and **B**reaking bonds = **ENDO**thermic (Figure C5.09).

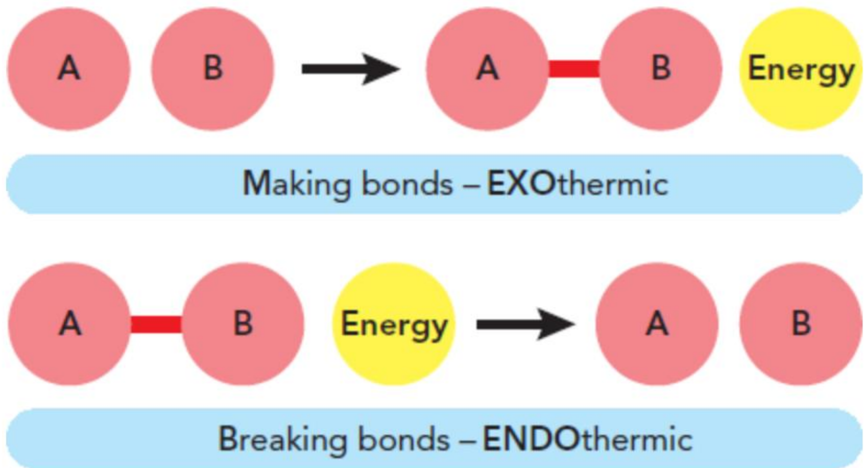

Figure C5.09 Memory aid for energy changes when making and breaking bonds: MEXOBENDO.

REFLECTION

How helpful do you find the memory aids such as 'EXothermic means that heat EXits the reaction; ENdothermic means that heat ENters the reaction' and 'MEXOBENDO'? Do you have a specific way to remember how to draw an energy level diagram? What strategies could you use to help you learn the ideas about chemical energetics covered in this chapter?

Activation energy

Although the vast majority of reactions are exothermic, only a few are totally spontaneous and begin without help at normal temperatures (e.g. sodium or potassium reacting with water). More usually, additional energy is required to start the reaction. When fuels are burnt, for example, energy is needed to ignite them to start the reaction (Figure C5.10). This energy may come from a spark, a match or sunlight. It is called the **activation energy (E_a)**. It is required because initially some bonds must be broken before any new bonds can be formed. Once the reaction has begun, the energy released as new bonds are formed allows the reaction to continue.

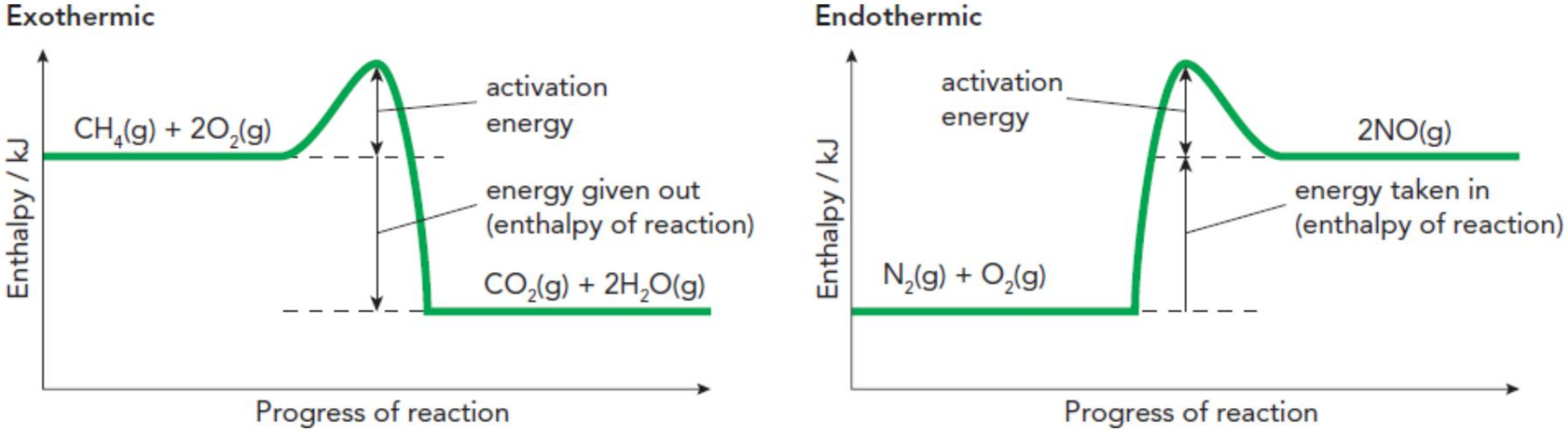

Figure C5.10 Energy level diagram for **a:** the burning of methane, and **b:** the formation of NO showing the need for activation energy to start the reaction.

Questions

C5.04 Is bond breaking an endothermic or an exothermic process?

C5.05 Hydrogen peroxide decomposes to produce water and oxygen. The equation is:

$$2H_2O_2(g) \rightarrow 2H_2O(g) + O_2(g)$$

Draw an energy level diagram for this reaction (there is a small amount of energy needed for this reaction to start).

C5.06 Explain why every chemical reaction must have an activation energy, even though for some reactions it may have a low value.

ACTIVITY C5.01

The chemical roller coaster

Have you noticed that energy level diagrams (reaction profiles) look a little bit like a roller-coaster ride? They go steeply up (taking in energy from the surroundings) and then there is a peak before they come racing down (releasing energy back into the surroundings as they do).

Using the exothermic reaction between hydrogen and oxygen, create a cartoon strip that explains the main stages in the energy level diagram (see Figure C5.11).

Think of the hydrogen and oxygen molecules being split into atoms before recombining to form water molecules.

Your cartoon strip should include:

- the key scientific words: activation energy, enthalpy change, exothermic, bond breaking, bond making
- at least five diagrams: start with just the reactants, then show the process of bond breaking, the activation energy (peak), the process of bond making and end with the products.

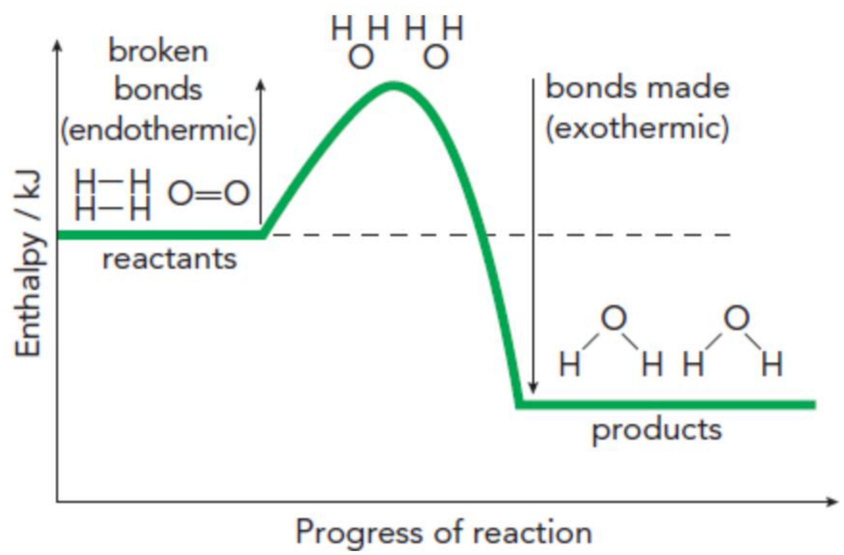

Figure C5.11 Energy level diagram for the reaction between hydrogen and oxygen.

You might want to see the reaction from the view of one of the atoms involved, giving them the stage to explain what is happening.

Peer assessment